

Model Regularization

Overfitting, Bias-variance decomposition, L1 and L2 regularization

Machine Learning and Data Mining, 2026

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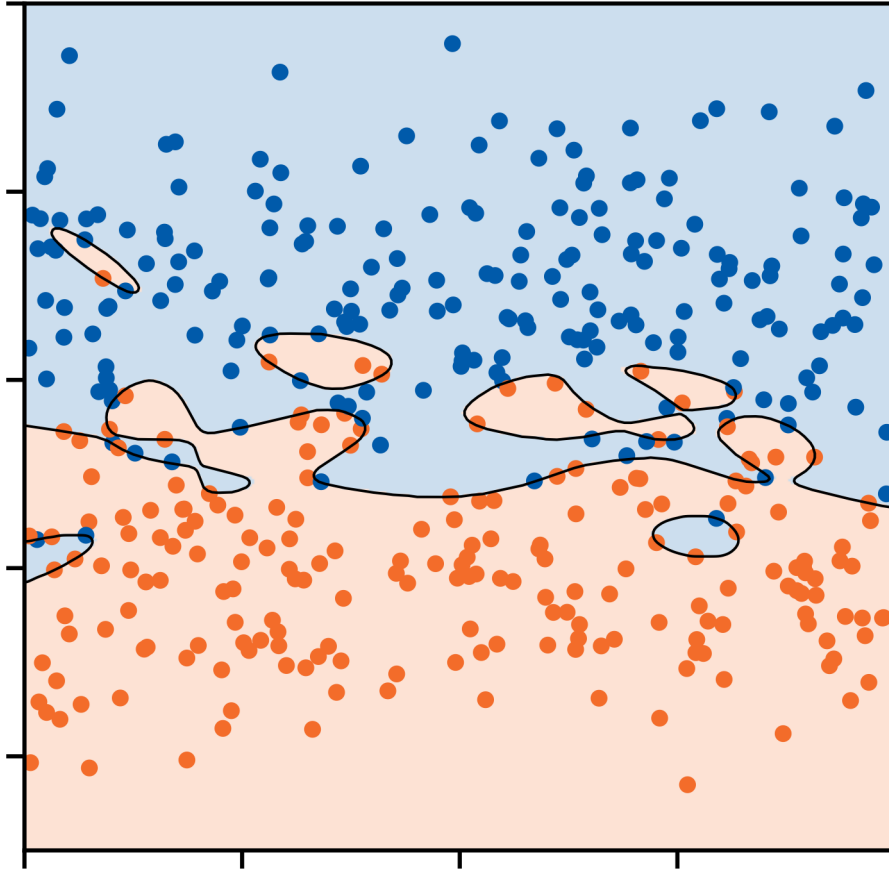


September 16, 2026

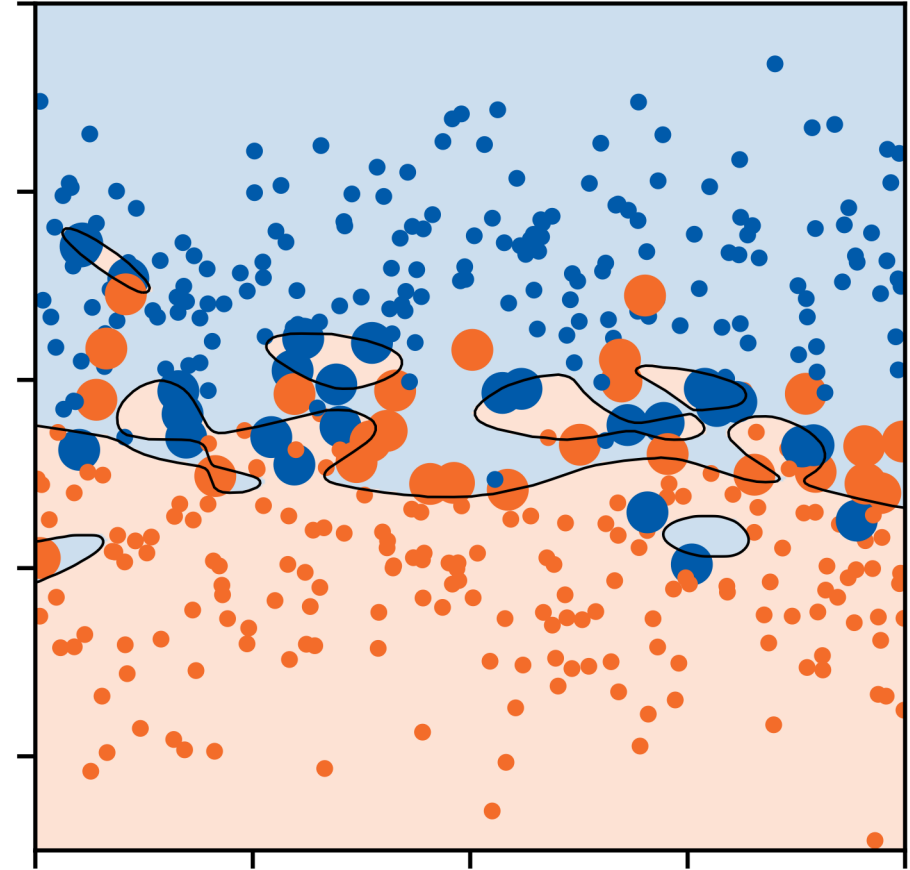
The problem of overfitting



Overfitting in classification



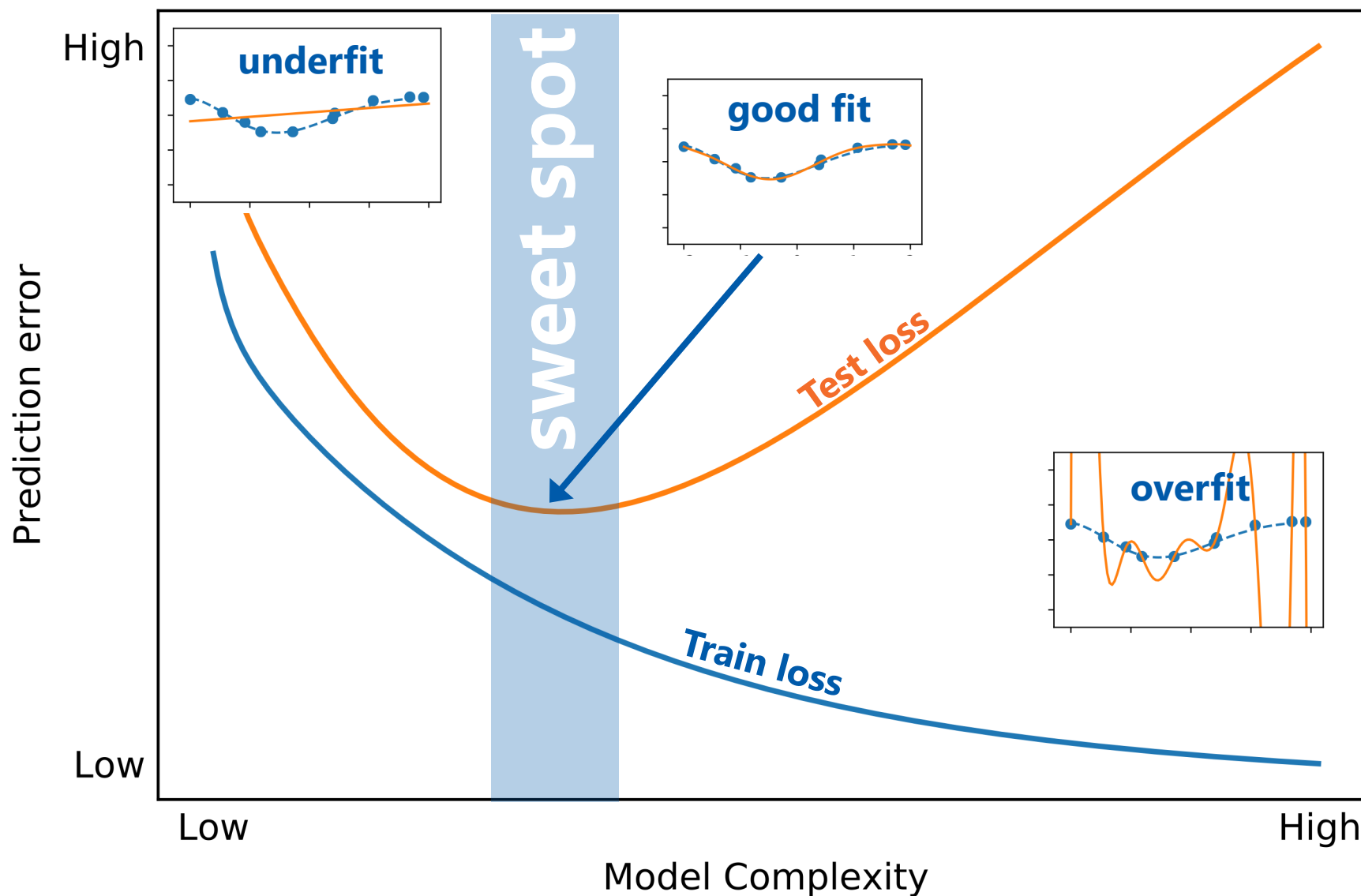
Training set



Test set

Large points =
classification error

How to check whether a model is good?



Check the loss on the **test data** – i.e. data that the learning algorithm hasn't seen

The goal is to find the **right level of limitations** – not too strict, not too loose

Prediction error decomposition



Prediction error decomposition

Assume there's the following (unknown) **relation between the features and targets**

$$y = f(x) + \varepsilon$$

where ε is some random noise:

$$\mathbb{E}[\varepsilon] = 0$$

$$\mathbb{D}[\varepsilon] = \sigma_{\varepsilon}^2$$

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Let's denote our training set as τ .

We want to study the **expected squared error** for the model $\hat{f}_\tau$ trained on it:

$$\text{exp. sq. err}(x) = \mathbb{E}_{\tau, y|x} \left[(\hat{f}_\tau(x) - y)^2 \right]$$

Prediction error decomposition

$$\begin{aligned}\text{exp. sq. err}(x) &= \mathbb{E}_{\tau, y|x} \left[(\hat{f}_{\tau}(x) - y)^2 \right] \\ &= \mathbb{E}_{\tau, y|x} \left[\left(\hat{f}_{\tau}(x) - \mathbb{E}[y|x] + \mathbb{E}[y|x] - y \right)^2 \right]\end{aligned}$$

Prediction error decomposition

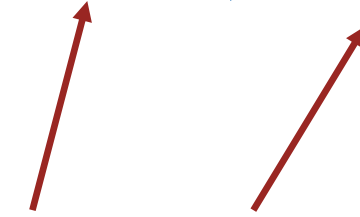
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**Prediction of the
"expected model"**

Prediction error decomposition

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**Ground truth
(without the
noise)**



Prediction error decomposition

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(grouping the terms, then expanding the square)

Prediction error decomposition

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(easy to show that all the cross term expectations are 0)

$$= \mathbb{E}_{\tau} \left[\left(\hat{f}_{\tau}(x) - \mathbb{E}_{\tau'}[\hat{f}_{\tau'}(x)] \right)^2 \right] + \left(\mathbb{E}_{\tau'}[\hat{f}_{\tau'}(x)] - f(x) \right)^2 + \mathbb{E}_{y|x} [(f(x) - y)^2]$$

**Variance of the
model**

i.e. how “unstable” the model is wrt
the noise in the training data

Prediction error decomposition

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how much the “expected model”
differs from the ground truth



Squared bias

Prediction error decomposition

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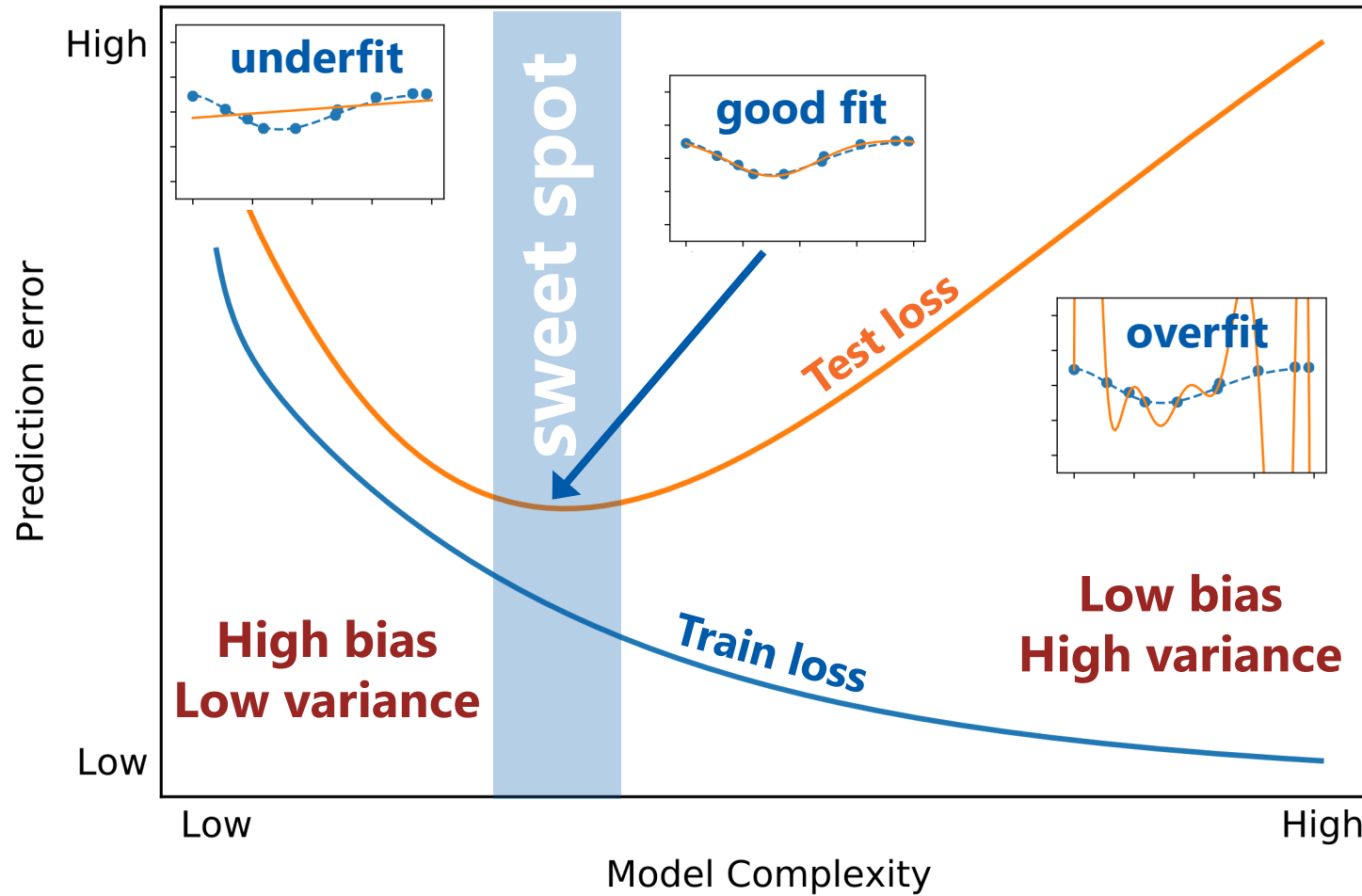
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**Irreducible
error**

$$(\quad = \mathbb{E}[\varepsilon^2] = \sigma_{\varepsilon}^2)$$

Bias-variance tradeoff



Typically there's a **tradeoff** between the two sources of error

Example: bias and variance of a linear model

Bias and variance error components can be calculated analytically for linear models

Simplification:

for each expectation term $\mathbb{E}_{\tau}$ let's consider **the features fixed**, i.e. $X_{\tau} \equiv X$ (the design matrix is constant), and only the **target vector y_{τ} is random**)

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Simplification:

for each expectation term $\mathbb{E}_\tau$ let's consider **the features fixed**, i.e. $X_\tau \equiv X$ (the design matrix is constant), and only the **target vector y_τ is random**)

Recall the solution for the linear regression model with the MSE loss:

$$\hat{f}_\tau(x) = \theta_\tau^\top x = x^\top \theta_\tau$$

$$\theta_\tau = (X^\top X)^{-1} X^\top y_\tau$$

Example: bias and variance of a linear model

Let's look at the **bias term** from the error decomposition:

$$\text{bias}(x) = \mathbb{E}_{\tau}[\hat{f}_{\tau}(x)] - f(x)$$

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$$\text{bias}(x) = \mathbb{E}_{\tau}[\hat{f}_{\tau}(x)] - f(x) = \mathbb{E}_{\tau} \left[x^T (X^T X)^{-1} X^T y_{\tau} \right] - x^T \theta_{\text{true}}$$

We'll also assume that
the **true dependence**
is linear indeed



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I.e. linear regression model is **unbiased**
as long as the true dependence is linear

Example: bias and variance of a linear model

Now let's look at the **variance term**:

$$\text{variance}(x) = \mathbb{E}_{\tau} \left[\left(\hat{f}_{\tau}(x) - \mathbb{E}_{\tau'} [\hat{f}_{\tau'}(x)] \right)^2 \right]$$

It can then be shown that:

$$\text{variance}(x) = \sigma_{\varepsilon}^2 x^T (X^T X)^{-1} x$$

So the variance error component is a **quadratic form**, defined by the $(X^T X)^{-1}$ matrix.

[derivation]

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$$\text{variance}(x) = \mathbb{E}_{\tau} \left[\left(\hat{f}_{\tau}(x) - \mathbb{E}_{\tau'} [\hat{f}_{\tau'}(x)] \right)^2 \right]$$

Note that $\hat{f}_{\tau}(x)$ can be thought of as a **linear transformation** to the training targets vector y_{τ} :

$$\hat{f}_{\tau}(x) = x^T \theta_{\tau} = x^T (X^T X)^{-1} X^T y_{\tau} = h^T(x) y_{\tau}$$

$$h^T(x) = x^T (X^T X)^{-1} X^T$$

[derivation]

$$\begin{aligned}\text{variance}(x) &= \mathbb{E}_{\tau} \left[\left(h^T(x) y_{\tau} - \mathbb{E}_{\tau'} [h^T(x) y_{\tau'}] \right)^2 \right] = \mathbb{E}_{\tau} \left[\left(h^T(x) \left(y_{\tau} - \mathbb{E}_{\tau'} [y_{\tau'}] \right) \right)^2 \right] \\ &= \mathbb{E}_{\tau} \left[h^T(x) \left(y_{\tau} - \mathbb{E}_{\tau'} [y_{\tau'}] \right) \left(y_{\tau} - \mathbb{E}_{\tau'} [y_{\tau'}] \right)^T h(x) \right] \\ &= h^T(x) \mathbb{E}_{\tau} \left[\left(y_{\tau} - \mathbb{E}_{\tau'} [y_{\tau'}] \right) \left(y_{\tau} - \mathbb{E}_{\tau'} [y_{\tau'}] \right)^T \right] h(x) \\ &= h^T(x) \text{cov}_{\tau} [y_{\tau}, y_{\tau}] h(x) = \sigma_{\varepsilon}^2 h^T(x) h(x)\end{aligned}$$

[derivation]

$$\text{variance}(x) = \sigma_{\varepsilon}^2 h^T(x) h(x)$$

$$= \sigma_{\varepsilon}^2 x^T (X^T X)^{-1} X^T X (X^T X)^{-1} x$$

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Example: bias and variance of a linear model

We can diagonalize $X^T X$:

$$\text{variance}(x) = \sigma_\varepsilon^2 x^T (X^T X)^{-1} x = \sigma_\varepsilon^2 \tilde{x}^T \Lambda^{-1} \tilde{x}$$

where $\Lambda = \text{diag}\{\lambda_1, \dots, \lambda_d\}$ is the matrix of eigenvalues of $X^T X$.

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This means that **small eigenvalues amplify the model variance**.

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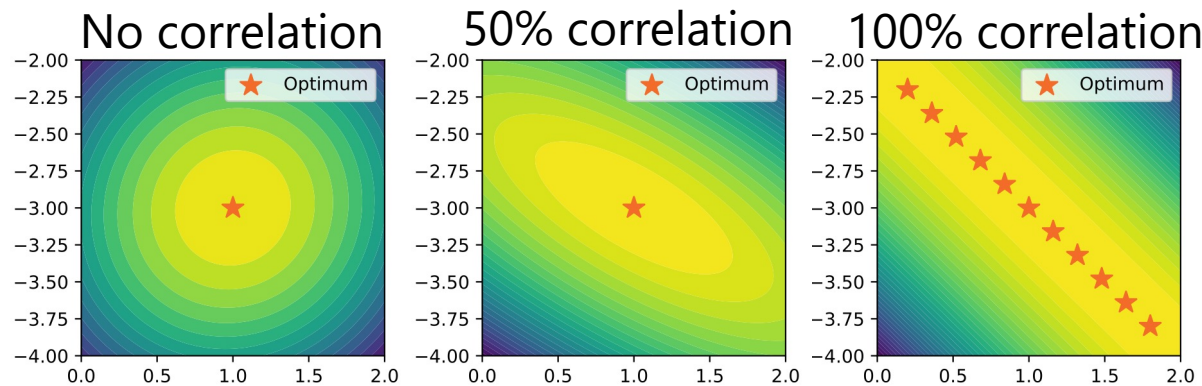
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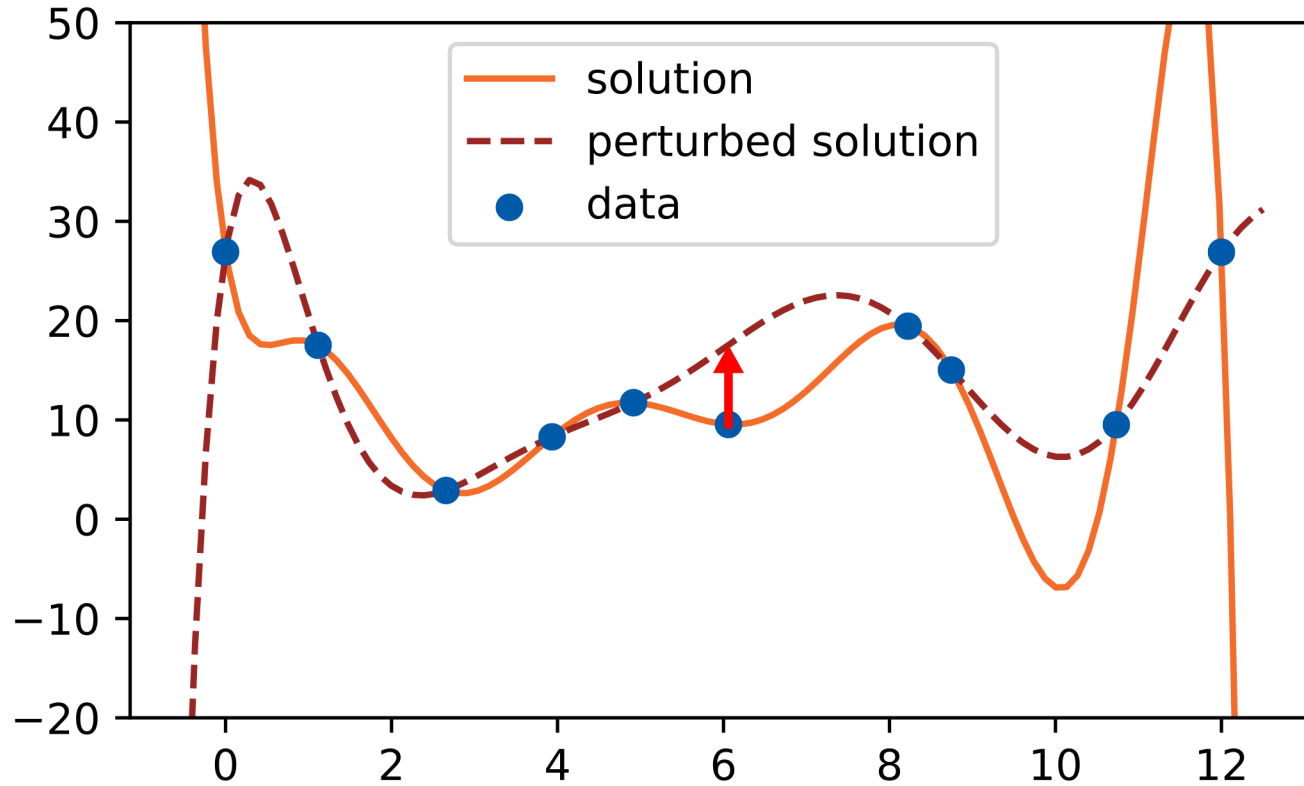
This means that **small eigenvalues amplify the model variance**.

This happens when $X^T X$ is ill-defined e.g. when the features are correlated



MSE loss values
as a function
of model parameters

High-variance model



Small perturbation in data



Large change in prediction

Regularization



How can we reduce the variance?

If only we could **increase the eigenvalues** of $X^T X$...

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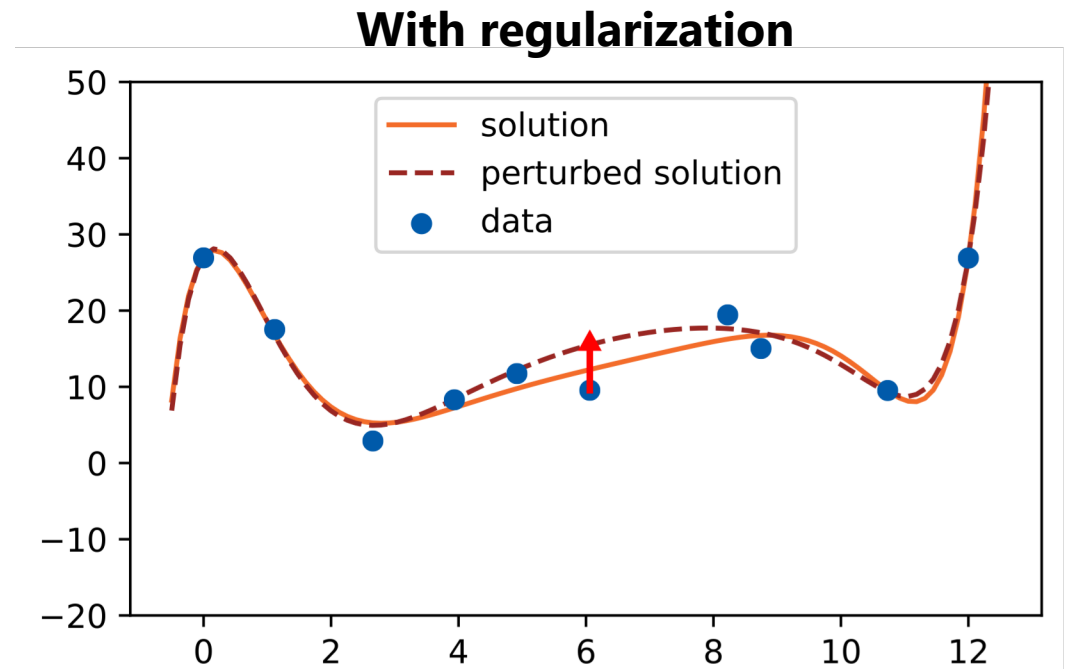
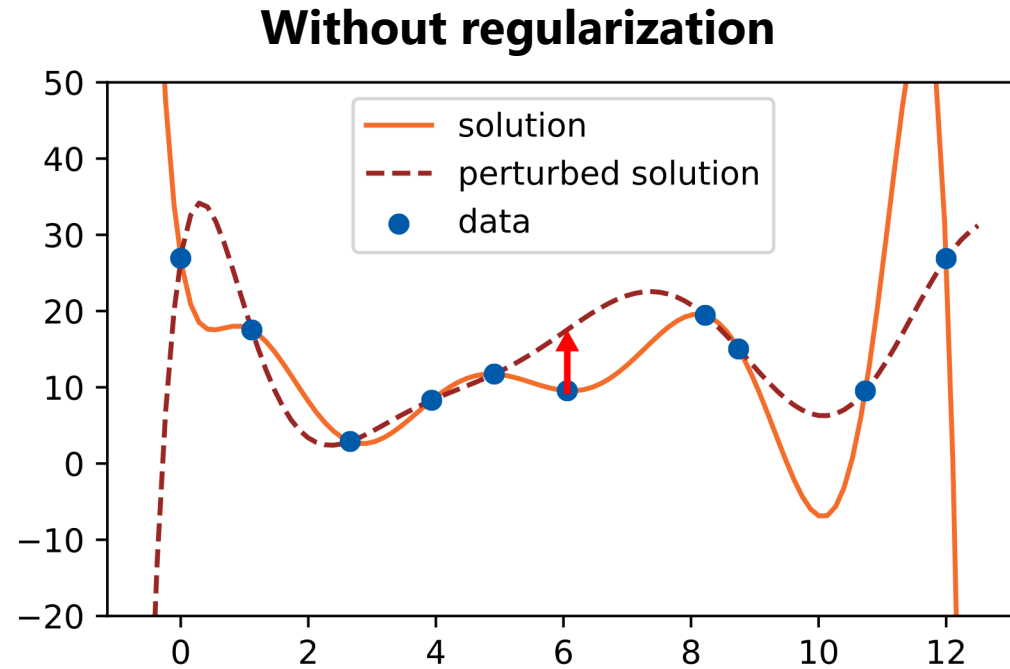
$$X^T X \rightarrow X^T X + \alpha I,$$

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I.e. we are **changing the solution** to:

$$\hat{f}_\tau(x) = x^T (X^T X + \alpha I)^{-1} X^T y_\tau$$

The effect of regularization



Note: the regularized model is **no longer unbiased!**

I.e. we **increased bias to reduce variance**

What problem did we solve?

We have the solution:

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Let's reverse engineer the loss function it optimizes:

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$$X^T (X \theta_\tau - y_\tau) + \alpha \theta_\tau = 0$$

In fact this is the $\partial/\partial\theta_\tau \mathcal{L} = 0$ equation for:

$$\mathcal{L} = \|X \theta_\tau - y_\tau\|^2 + \alpha \|\theta_\tau\|^2$$

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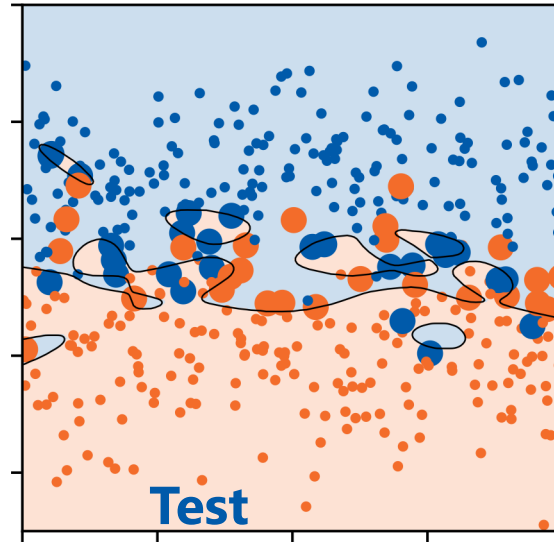
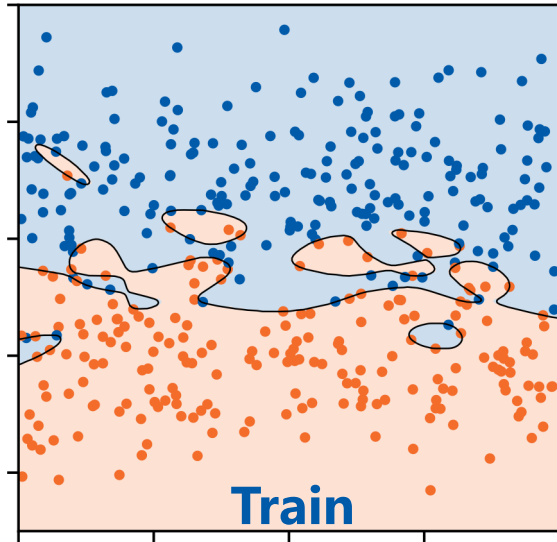
In other words, this linear model:

$$\hat{f}_\tau(x) = x^T (X^T X + \alpha I)^{-1} X^T y_\tau$$

minimizes **MSE loss** with **L2 penalty term** on the model parameters.

Such model is also called
ridge regression

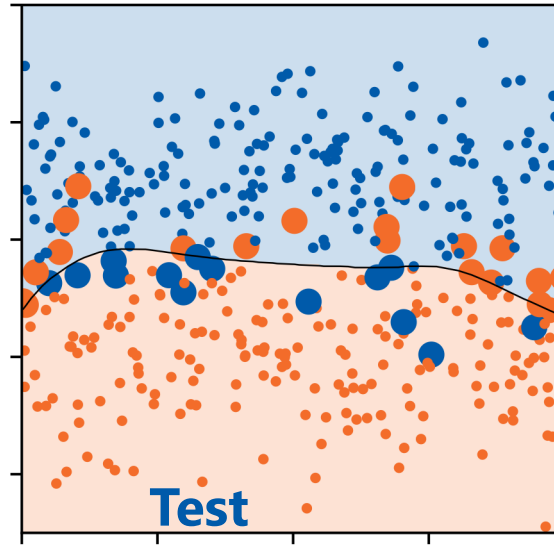
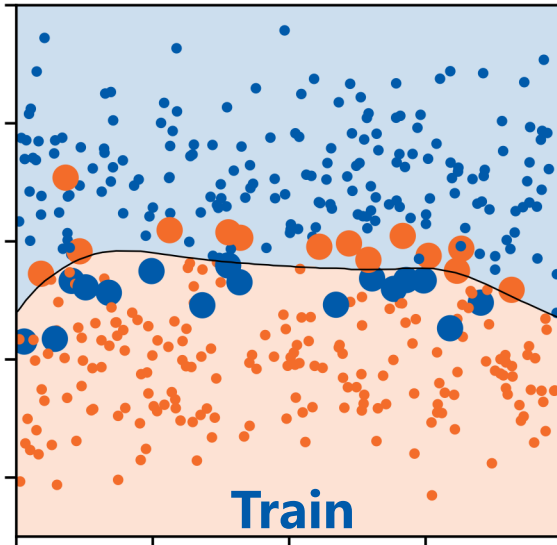
Example: L2-regularized classification



Without regularization

By regularizing the model we **increase the train loss** and **decrease the test loss**

This improves the **generalizability** of the model



With regularization

Various regularization methods

L2 regularization (Ridge):

$$\mathcal{L} = \|X\theta_\tau - y_\tau\|^2 + \alpha\|\theta_\tau\|^2$$

L1 regularization (Lasso):

$$\mathcal{L} = \|X\theta_\tau - y_\tau\|^2 + \alpha\|\theta_\tau\|_1$$

Elastic net:

$$\mathcal{L} = \|X\theta_\tau - y_\tau\|^2 + \alpha\|\theta_\tau\|^2 + \beta\|\theta_\tau\|_1$$

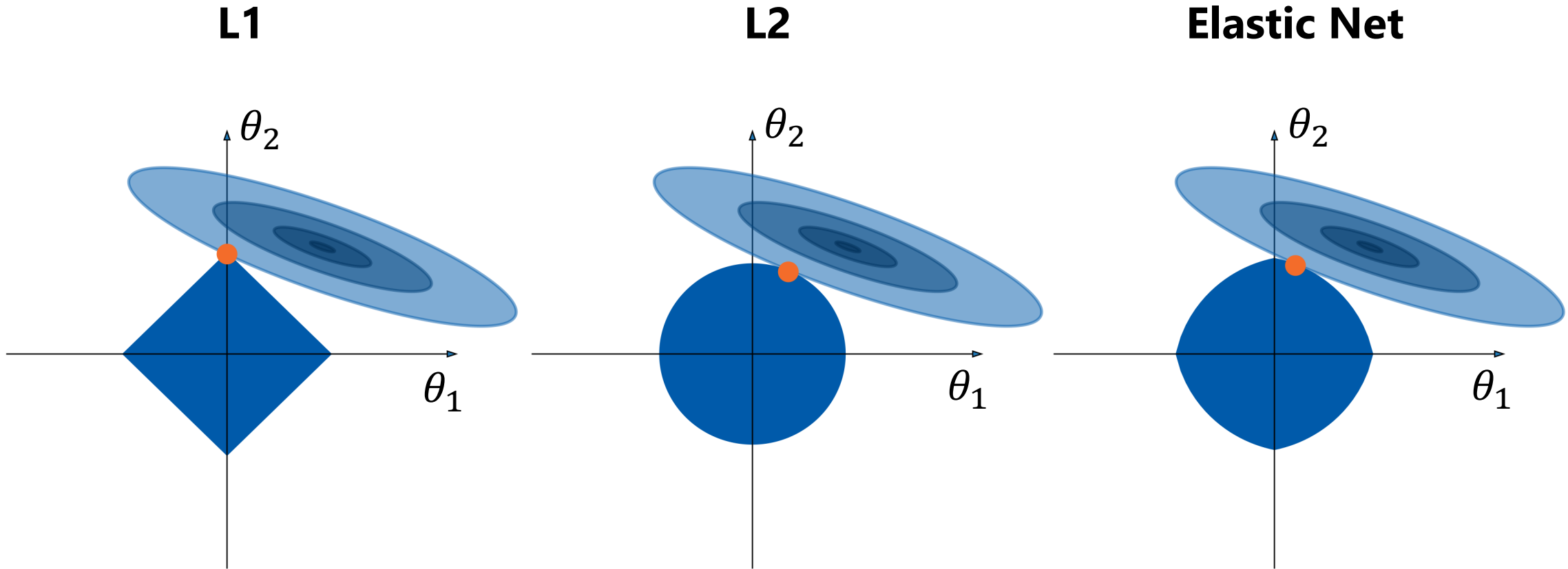
L2 norm:

$$\|x\|^2 \equiv \sum_{i=1\dots d} x_i^2$$

L1 norm:

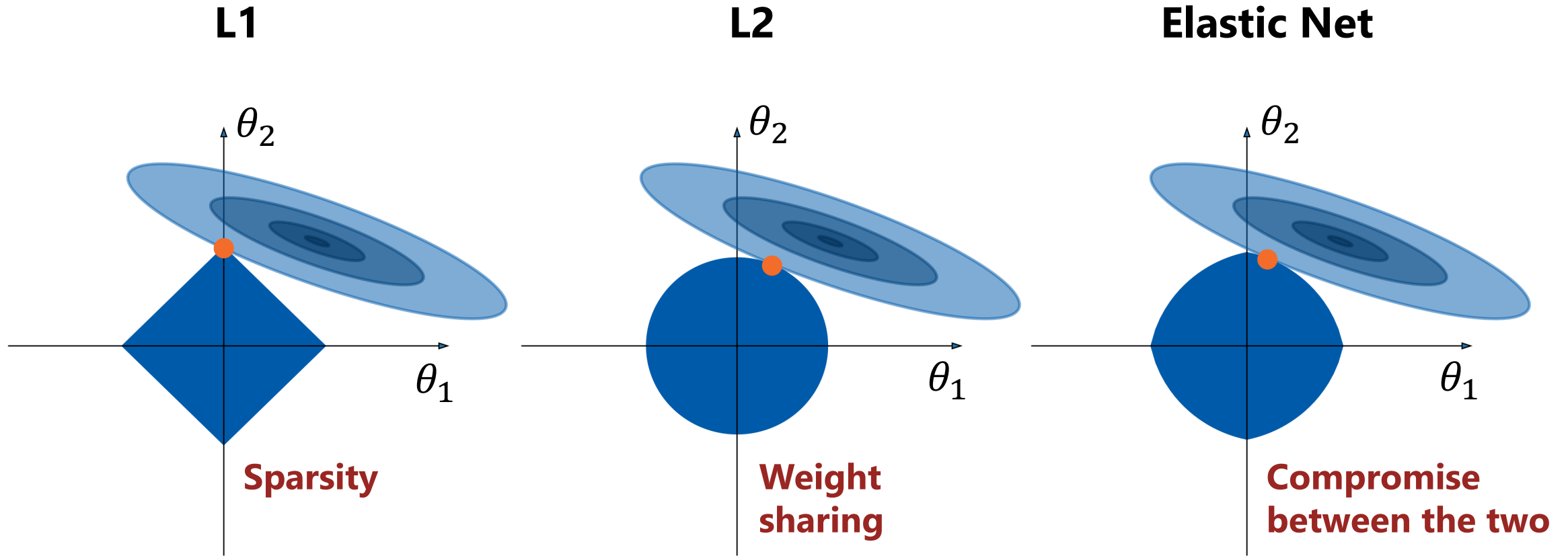
$$\|x\|_1 \equiv \sum_{i=1\dots d} |x_i|$$

Properties of different regularization methods



They all drive the weights towards **smaller values**
Yet they **induce different properties** of the solution

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Linear regression is **unbiased**, while its variance is large when $X^T X$ matrix is **ill-defined**

Typically regularization reduces the variance with the price of **increasing the bias**

Different regularization techniques induce different properties of the solution

Thank you!



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